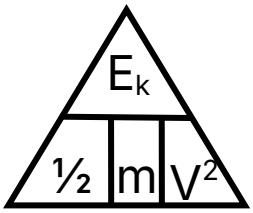
Calculating Kinetic Energy, E_k

The kinetic energy of a moving object is dependent on 2 factors; the mass (m) of the object & its velocity (v). The formula for calculating kinetic energy is:



$E_k = ½ mv^2$

Kinetic energy = ½ x mass x velocity²

- Mass of the object is measured in kg
- Velocity is measured in m/s or ms^{-1} (either one is accepted as they are the same unit of measurement)

Example

The total mass of a go-kart and the driver is 90kg. The go-kart is travelling at 40 m/s, how much kinetic energy does it possess?

What are my knowns?

E_k* = ½ *mv²

= ½ x 90 x 40²

= 0.5 x 90 x 1600

= 72,000 joules

= 72 kJ



Questions

(Calculate answer to two significant figures)

1. The first driver of a go-kart had a mass of 50kg and travelled a top speed of 40m/s. They were replaced by a driver with a mass of 70kg. Calculate the kinetic energy both drivers in the go-kart possess when travelling at the same speed. Comment on the difference you find and why it occurs.

2. A crane raises a load of 200kg to a height of 30m in one minute. Determine the kinetic energy of the load when it's moving.

3. A racing car drives around a circuit at a constant speed of 50m/s. If the car has a kinetic energy of 500kJ, what is the mass of the car?

4. A plane has a mass of 9000kg. It flies at a speed of 100m/s. Calculate the kinetic energy of the plane.

5. A runner crosses the finishing line at a speed of 7m/s. At this point the runner has a kinetic energy of 1862J. Calculate the mass of the runner.

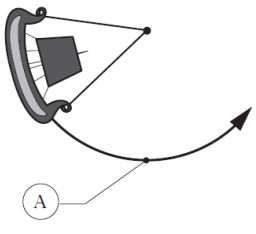
6. A remote controlled car has a speed of 1.2m/s. It has a kinetic energy of 0.216J. Calculate the mass of the car.

7. A lorry of mass 2800kg is travelling at a constant speed. It has a kinetic energy of 132kJ. Calculate the speed of the lorry.

8. A motorbike has a mass of 250kg and is travelling at 18m/s. A car has a mass of 950kg and travels at 12.5m/s. Calculate which has the greater kinetic energy.

9. A car of mass 850kg is travelling at a speed of 20m/s. The brakes are applied and the car reduces to a speed of 12m/s. Calculate the change is the car's kinetic energy.

10. A fairground swing is shown.



The swing and passengers have a combined mass of 2500kg.

(a) Calculate, showing all working and units:

(i) The kinetic energy at its lowest point when the velocity is 15 m/s;

(ii) The maximum height the swing will reach. (Assume no loss of energy). Note: You must use the potential energy equation to solve this problem: $E_p=mgh$. The next theory page explains calculating E_p.

(b) In reality, there will be some loss of energy in the swing.

(i) State one possible cause of energy loss.

(ii) State how this energy loss could be reduced.